Dear Hiring Manager,

I am excited to apply for the Associate AI/ML Engineer position at Optum. I am currently completing my MSc in Computer Science from the prestigious Sapienza University of Rome, specializing in Artificial Intelligence and Machine Learning. My background combines AI model development, backend engineering, and building practical AI-powered applications, which aligns strongly with Optum's mission of using technology to improve healthcare outcomes.

During my master's program, I have worked on developing and evaluating machine learning and deep learning systems across computer vision and NLP applications. My research on Continual Learning for Computer Vision involved evaluating Vision Transformer models, designing reproducible ML pipelines, and analyzing model performance under changing data conditions. I have also contributed to an IEEE conference publication on machine learning approaches for environmental prediction.

Beyond research, I have hands-on experience building end-to-end AI applications. I developed DocDelta, an AI-powered document intelligence platform using transformer-based embeddings, FastAPI, Docker, and AWS deployment services. The project involved designing REST APIs, integrating ML models into backend workflows, and deploying scalable AI services. I also developed PosePal, a real-time computer vision application that uses CNN-based classification for live webcam-based yoga pose recognition.

I have experience working with Python, PyTorch, TensorFlow, FastAPI, REST APIs, Docker, AWS, SQL, and software development practices. Additionally, my background includes Java, frontend development fundamentals, databases, and collaborative software development workflows. I am eager to continue expanding my skills while contributing to AI-enabled applications and cloud-native solutions at Optum.

I am particularly interested in Optum because of the opportunity to apply artificial intelligence and software engineering to real-world healthcare challenges that directly impact people's lives. I would welcome the opportunity to contribute my technical skills, curiosity, and problem-solving approach to your team.

Thank you for your consideration. I look forward to the opportunity to discuss how my experience aligns with this role.

Sincerely,

Swati Aggrawal